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Research paper

Profile and contact force estimation of cable-driven continuum robots in presence of obstacles

K.P. Ashwin[a], Soumya Kanti Mahapatra[b], Ashitava Ghosal[b,*]

[a] Department of Mechanical Engineering, BITS Pilani K.K. Birla Goa Campus, Goa, 403726, India
[b] Department of Mechanical Engineering, Indian Institute of Science, Bangalore, 560012, India

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ABSTRACT

Accurate prediction of shape and contact forces significantly improves the performance of a continuum robot during its operation in obstacle-laden environments. This paper presents an optimization-based mathematical framework to predict the bending profile of a cable-driven continuum robot in presence of obstacles. The kinematics model is derived from the concept of strain energy minimization and can easily incorporate obstacles as inequality constraints in the optimization-based approach. The location of point of contact can be identified by observing the Lagrange multipliers of the inequality constraints. Using the kinematics model and the principle of virtual work, a method to estimate the reaction forces at contact is proposed. The model shows high accuracy, with RMS error of 1.35 mm in prediction of the pose for experiments conducted on a 180 mm long robot prototype. Validation experiments are also conducted on the prototype by imposing contact at different locations on the robot. In all cases, the average error in predicting the contact force is found to be less than 1.0 g for applied loads ranging from 50 to 350 g.

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1. Introduction

A continuum robot, as opposed to rigid link robot is characterized by a continuous elastic structure which is deformed using different actuation techniques [1]. Continuum robots are convenient for situations where compliance is preferred such as during human-robot interactions or handling delicate objects. Due to the hyper-redundancy in the continuum robot, they are also suitable for operations in inaccessible areas as well as for obstacle avoidance problems. A class of continuum robots which are actuated using cables (or tendons) are of particular interest due to their simplicity in design and miniaturizability [2,3]. A typical cable-driven continuum robot (CCR) consists of a slender elastic *backbone* on which a set of disks are fixed with constant spacing (refer Fig. 1(a)). Through the holes in the disks, cables are routed along the length of the backbone and are terminated at the end-disk. By pulling the cable from the base, a moment arm is generated at the tip, causing the CCR to bend.

Accurate mathematical models for the physics of CCR is of great interest since they can be used in robot design and control. A good mathematical model for CCR also helps to understand the behavior of the robot during contact with its environment. Since CCRs are generally light-weight and their applications involve slow movements, study of statics and kinematics are usually prioritized over dynamics. A detailed review of such analyses can be found in the works of Webster III

* Corresponding author.
E-mail addresses: ashwinp@goa.bits-pilani.ac.in (K.P. Ashwin), soumyam@iisc.ac.in (S.K. Mahapatra), asitava@iisc.ac.in (A. Ghosal).

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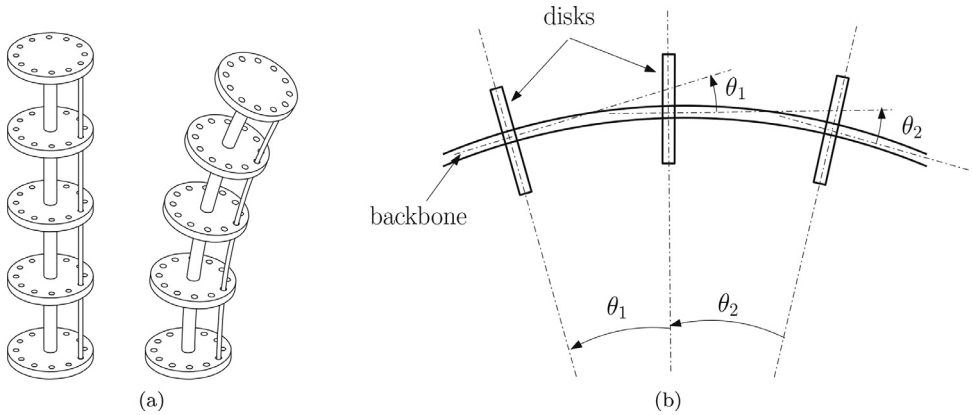


Fig. 1. (a) Cable-driven continuum robot with cables terminating at the tip. (b) Slope angle is equivalent to the change in angle between consecutive disks of CCR.

and Jones [4], Chirikjian [5], Rao et al. [6] etc. In [6], the authors classified the modeling approaches as distributed and lumped models based on the type of backbone parametrization and their modeling assumptions. In distributed models, the backbone is considered as a 3-D curve which is represented using a single parameter, usually the normalized arc-length of the curve. For such models, the kinematic analysis is conducted by representing the rate of change of position and rotation of backbone in the Frenet-Serret frame as functions of strain variables [4]. For lumped model analysis, the continuum backbone is discretized into finite segments and variables such as Denavit-Hartenberg parameters or joint co-ordinates are used for parameterizing the backbone curve [7,8,9].

For static analysis of distributed models, the forces and moments distributed in the entire length of the robot are first accounted for. Different modeling considerations such as cable routing characteristics [10] and additional forces [11] add different levels of complexity to these formulations. For straight-routed robots, Euler-Bernoulli beam theory is used for constitutive relations [12] while for generally-routed robots, Cosserat-rod theory is used [10]. The solution to these boundary-value problems are usually computed by numerical integration. For static analysis of lumped models, a pseudo-rigid body approach is commonly used with the backbone considered as a series of rigid links connected using torsional springs [13], [14]. The internal torques of the backbone segments is related to the applied torques (and forces) by means of the robot's Jacobian where the stiffness of springs and the optimal lengths of segments are determined from experiments. In the distributed models, external force constraints are incorporated as boundary conditions for the governing differential equations [15]. In the lumped model, the kinematic and static constraints are directly applied by specifying the position or load at the individual segments [14].

Understanding the characteristics of the robot during contact with the environment is essential for their control in obstacle-laden environments. In the above cases, prior knowledge of the location of contact between the CCR and the object is necessary to estimate its post-collision characteristics so that either the differential equations or the solution procedure can be modified accordingly. In order to estimate the exact location of contact and the reaction forces due to the same, different strategies are used in the literature. In [16], the authors present an algorithm to estimate the contact location by reducing the deviation between the theoretical and measured values of loci of the fixed centroid of the end-disk. This is further extended in [17] by minimizing the deviation in instantaneous screw axis of the motion. Many researchers use sensors which can directly measure the robot profile and estimate the contact characteristics with the help of a static model of the robot [18,19,20,21]. Using explicit sensors, the contact forces can be obtained as shown in [14], but the feasibility of such an approach depends on the design specifications of the robot. In [22], the authors use a model-based hybrid controller to estimate the contact characteristics while in [23] a model-based controller uses Extended Kalman Filter to estimate the forces. Finite element method (FEM) based models using Cosserat-rod theory and Euler-Bernouilli beam theory for CCR modelling and contact analysis are shown to give results with good accuracy [24,25,26]. Apart from linear elastic considerations, hyper-elastic models are also used for modelling continuum robots as can be seen in [27].

Except for the FEM-based models, all the above cases of contact characteristics estimation require employment of one or more sensing techniques and advanced control algorithms to estimate the exact location of contact and the reaction forces due to collision. In the following sections, we present an optimization-based framework for solving the kinematics of CCR in presence of obstacles where prior knowledge about the exact location of contact on the robot is not necessary. In the approach described in this paper, only the position and shape of the external obstacles need to be given, and the profile of the robot after collision can be calculated without the use of any external sensors. The framework is an improved version of the model developed in [28], where the CCR is parametrized as a lumped model comprising of a series of adjoined hypothetical 4-bar linkages. The accuracy of predicting the pose using this method is verified from experiments and it is also demonstrated that the exact contact location on the robot can be obtained as a by-product of the solution procedure. A method to estimate the contact forces on obstacles, using the developed kinematics model and the principle of virtual-

Table 1
Nomenclature.

Symbol	Description
B	Flexural rigidity
ϕ	Slope angle
s	Arc-length parameter
L_0	Length of backbone
l_0	Length of a backbone segment
n	Number of backbone segments
a	Radial distance between the centre of the backbone and the cables
$\mathbf{X}_0$	Position vector of undeformed backbone at the disk
$\mathbf{X}_a$	Position vector of the first unactuated cable at the disk
$\mathbf{x}_0$	Position vector of the deformed backbone at the disk
$\mathbf{x}_a$	Position vector of the first actuated cable at the disk
$\mathbf{X}_b$	Position vector of the second unactuated cable at the disk
$\mathbf{x}_b$	Position vector of the second actuated cable at the disk
l	Length of cable pulled
$\mathbf{f}(\mathbf{x})$	Function for boundaries of obstacles
x_e, y_e	X and Y co-ordinates of the tip of the robot
θ_e	Orientation of the robot tip
P_x, P_y	Reaction forces in X and Y directions, at the tip
M_e	Moment at the tip
R_x, R_y	Reaction forces at the base
M_r	Reaction moment at the base
$\delta x, \delta y$	Virtual displacements in X and Y directions
$\delta\theta$	Virtual displacement in θ direction
F_l	Applied load
μ	Coefficient of friction
d	Distance between backbone and the obstacle(load cell)
l_a, l_b	Length of the first and second cable inside a segment

work, is also presented in this work. The method is experimentally verified and it is shown that the contact force can be estimated with average error less than 1.0 g. The demonstrated method will be advantageous for applications where intrinsic or extrinsic force measurement is not practically viable.

The rest of the paper is organized as follows: In Section 2, the optimization-based kinematics formulation starting from strain energy minimization principles is introduced first. The extension of kinematics formulation to include obstacles and identification of contact location is presented next. Then contact force estimation using virtual work principle is shown and lastly, the extension of the formulation to 3D is presented. In Section 3, the numerical results from the optimization-based formulation are compared with the results obtained from experiments conducted on a 3D-printed prototype. This is followed by a discussion on the computation time and accuracy. In Section 4, the main conclusions and the scope for future extensions of this work are presented.

2. Methodology

2.1. Optimization-based kinematics of CCR

In this section, we derive the optimization-based kinematics of CCR from the principle of strain energy minimization. For the kinematic analysis shown in the sections 2.1, 2.2 and 2.3, we consider a CCR actuated using only one cable which is routed parallel to the backbone. Description of the symbols which will be used in the rest of this paper is given in Table 1.

In most of the CCR designs, the backbone is a long slender member whose deformation can be considered as an elastic beam undergoing bending deformation. Upon application of an external load, the member deforms such that its strain energy is minimum at the static equilibrium. This is equivalent to the expression

$$\min_{\phi} \int_0^{L_0} B\left(\frac{d\phi}{ds}\right)^2 ds \tag{1}$$

where ϕ is the slope angle, s is the arc-length parameter, B is the flexural rigidity and L_0 is the length of the backbone. In a finite difference form, the above equation can be written as

$$\min_{\Delta\phi} \sum_{i=1}^{n} B\left(\frac{\Delta\phi}{\Delta s}\right)^2 \delta s \tag{2}$$

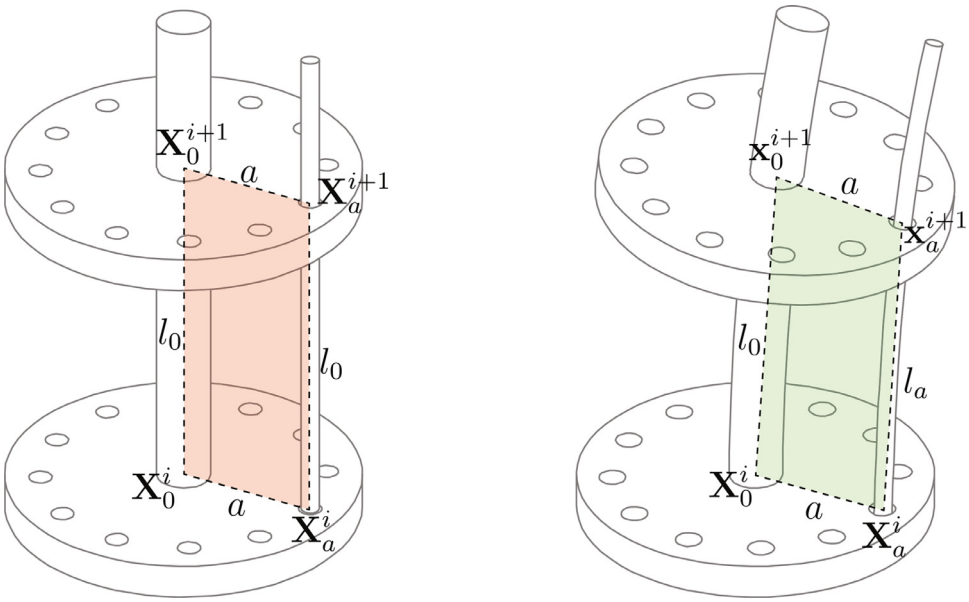


Fig. 2. Nomenclature and variables used in optimization.

Assuming that the backbone is discretized into equal number of segments and the material is isotropic, the above equation becomes

$$\min_{\Delta\phi} C \sum_{i=0}^{n} (\Delta\phi)^2 \tag{3}$$

where C is a constant. Hence, the CCR in static equilibrium will have a deformed profile where the sum of squares of the change in slope angles is minimized – modulo a constant. Since the construction of CCR ensures that the disk is always perpendicular to the backbone, the change in slope angle is also equivalent to the angle between the two consecutive disks (refer Fig. 1(b)). Hence, minimization of the latter ensures that the kinematic criterion for the backbone with perpendicular disks is met. When the CCR is actuated, the above condition is augmented by the kinematic constraints imposed by the length of the cables in the segment and the size of the guiding disks. With these added constraints, we can uniquely identify the profile of the actuated CCR.

For mathematically representing the minimization problem and the constraints, the following nomenclature is assigned (refer Fig. 2):

- The backbone and cables are discretized into n segments, each comprising of a closed loop made of a backbone segment, top disk, cable segment and a bottom disk.
- The segment lengths of backbone and cable are l_0 and l_a respectively and they take the values $l_0 = l_a = L_0/n$ in the un-actuated state.
- Distance between the cable and the centre of the backbone is fixed along the disk and is given by the constant a which is the same for all the disks.

For analysis, the numbering of segments start from bottom, where the bottom most disk has the index $i = 0$, and the disk at the tip of the robot has the index $i = n$. The vertices of the loop in the ith segment are termed $\left(\mathbf{X}_0^i, \mathbf{X}_a^i, \mathbf{X}_a^{i+1}, \mathbf{X}_0^{i+1}\right)$ in the counter-clockwise direction starting from the bottom left vertex. When the CCR is actuated, the linkage reduces its length from $\frac{L_0}{n}$ to $\frac{L_0 - l}{n}$ where l is the amount by which the cable is pulled. The loop deforms as shown in Fig. 2 and the quantities $\mathbf{X}_{(\cdot)}^{i+1}$ change to $\mathbf{x}_{(\cdot)}^{i+1}$.

Using the aforementioned nomenclature, the optimization problem can be written as:

$$\underset{\mathbf{x}_0, \mathbf{x}_a}{\arg\min} \quad \sum_{i=0}^{n-1} \left(\arccos\left(\left(\frac{\mathbf{x}_0^{i+1} - \mathbf{x}_a^{i+1}}{\|\mathbf{x}_0^{i+1} - \mathbf{x}_a^{i+1}\|} \right) \cdot \left(\frac{\mathbf{x}_0^{i} - \mathbf{x}_a^{i}}{\|\mathbf{x}_0^{i} - \mathbf{x}_a^{i}\|} \right) \right) \right)^2$$

Subject to:

$$\begin{aligned} \|\mathbf{x}_0^i - \mathbf{x}_0^{i-1}\| &= l_0 \\ \|\mathbf{x}_a^i - \mathbf{x}_a^{i-1}\| &= l_a \\ \|\mathbf{x}_0^i - \mathbf{x}_a^i\| &= a, \qquad i = 1, 2, 3, \ldots, n \end{aligned} \tag{4}$$

Given data: l_0, l_a, a;
where the variables without segment index represent the arrays $\mathbf{x}_{(\cdot)} = [\mathbf{x}^1_{(\cdot)}, \mathbf{x}^2_{(\cdot)}, \ldots, \mathbf{x}^n_{(\cdot)}]$ etc.

The above optimization problem is solved as a global optimization problem with constraints. This approach allows us to add global constraints to the minimization routine and can be used for obstacle avoidance [29]. The numerical results and experimental validation of the numerical results are presented in Section 3.

2.2. Kinematics model with obstacles

For the constrained global optimization problem, obstacle avoidance can be achieved by adding extra constraints in addition to the equality constraints in (4) [29,30]. For obstacle avoidance, the problem can be reformulated as:

$$\underset{\mathbf{x}_0,\mathbf{x}_a}{\arg\min} \quad \sum_{i=0}^{n-1}\left(\arccos\left(\left(\frac{\mathbf{x}_0^{i+1}-\mathbf{x}_a^{i+1}}{\|\mathbf{x}_0^{i+1}-\mathbf{x}_a^{i+1}\|}\right)\cdot\left(\frac{\mathbf{x}_0^{i}-\mathbf{x}_a^{i}}{\|\mathbf{x}_0^{i}-\mathbf{x}_a^{i}\|}\right)\right)\right)^2$$

Subject to:

$$\begin{aligned}
\|\mathbf{x}_0^i-\mathbf{x}_0^{i-1}\| &= l_0\\
\|\mathbf{x}_a^i-\mathbf{x}_a^{i-1}\| &= l_a\\
\|\mathbf{x}_0^i-\mathbf{x}_a^{i}\| &= a, \qquad i=1,2,3,\ldots,n\\
\mathbf{f}(\mathbf{x}_0) &\succeq 0\\
\mathbf{f}(\mathbf{x}_a) &\succeq 0
\end{aligned} \tag{5}$$

Given data: $l_0, l_a, a, \mathbf{f}(\mathbf{x})$
where $\mathbf{f}(\mathbf{x})$ is the function for boundaries of the obstacles[1]. For example, if $\mathbf{f}(\mathbf{x}) = (\mathbf{x}-x_c)^2 + (\mathbf{y}-y_c)^2 - r_c^2$, then the discrete points of the backbone and the cable will always lie outside a circle with center (x_c, y_c) and radius r_c.

As mentioned in Section 1, this method does not require explicit knowledge of the location of point of contact in the CCR. In fact, the location of the point of contact can be found by observing the Lagrange parameters of the constrained optimization problem. Wherever the nodal co-ordinates violate the non-linear constraint, the Lagrange multipliers turn non-zero and hence can be used as an identifier to determine the nodes which are in contact.

2.3. Estimating contact forces using the kinematics model

In this section, we show that the results from the kinematics can be used to estimate the contact forces using the principle of virtual work.

The position and orientation of the tip of the robot (x_e, y_e, θ_e) can be represented as a function of the change in cable length:

$$x_e = f_x(l), \quad y_e = f_y(l), \quad \theta_e = f_\theta(l) \tag{6}$$

where the functions f_x, f_y and f_θ can be obtained from the kinematics model. The variational displacements will be

$$\delta x_e = f_x^{'}\delta l, \quad \delta y_e = f_y^{'}\delta l, \quad \theta_e = f_\theta^{'}\delta l \tag{7}$$

where $f^{'}_{(\cdot)} = \frac{df(\cdot)}{dl}$. When the tip of CCR is constrained due to external object, a set of reaction forces and moments (P_x, P_y, M_e) as well as (R_x, R_y, M_e) are generated at the tip and the base respectively (see Fig. 3).

Let $\delta x, \delta y,$ and $\delta\theta$ denote the virtual displacement field corresponding to the reaction forces at the base. The virtual displacement at the tip due to (P_x, P_y, M_e) will be $\left(\delta x + f_x^{'}\delta l, \delta y + f_y^{'}\delta l,\right.$ and $\left.\delta\theta + f_\theta^{'}\delta l\right)$, respectively. By invoking the principle of virtual work, we get

$$-R_x\delta x - R_y\delta y - M_r\delta\theta + P_x\left(\delta x + f_x^{'}\delta l\right) + P_y\left(\delta y + f_y^{'}\delta l\right) + M_e\left(\delta\theta + f_\theta^{'}\delta l\right) = 0 \tag{8}$$

Regrouping the terms,

$$(-R_x + P_x)\delta x + (P_y - R_y)\delta y + (-M_r + M_e)\delta\theta + \left(P_x f_x^{'} + P_y f_y^{'} + M_e f_\theta^{'}\right)\delta l = 0 \tag{9}$$

from which we obtain the static equilibrium equations:

$$\begin{aligned}
R_x = P_x, \quad R_y = P_y, \quad M_r = M_e\\
P_x f_x^{'} + P_y f_y^{'} + M_e f_\theta^{'} = 0
\end{aligned} \tag{10}$$

From the kinematics, we get the functions $f_x^{'}, f_y^{'},$ and $f_\theta^{'}$. The component $M_e = M_r$ is the moment generated at the tip of the robot due to the applied load F_l, i.e., $M_e = a \times F_l$ [31]. Hence, by knowing one of the force components F_x or F_y, the other can be calculated.

[1] The generalized inequality symbol $\succeq$ is used to represent vector inequalities

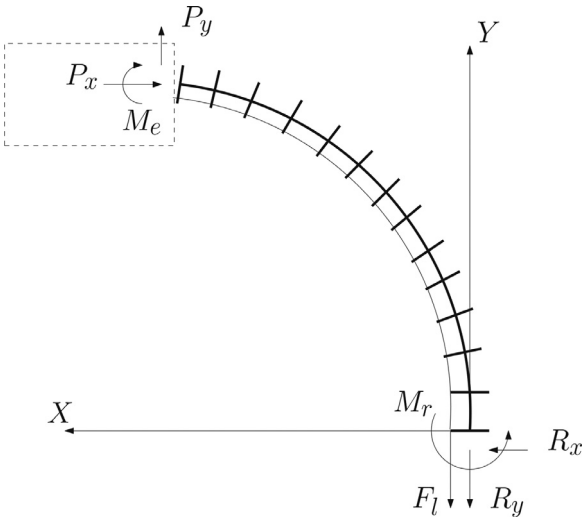


Fig. 3. Free body diagram of the CCR with obstacle in contact at the tip.

2.4. Extension to 3D

If two non-opposing cables are actuated simultaneously, the CCR bends in a plane which lies in between the two bending planes generated by pulling the two cables individually. The optimization-based kinematics formulation can be extended to such 3D cases by minimizing the slope angles in both the individual bending planes. This is equivalent to minimization of total strain energy which consists of two strain energy components corresponding to the two bending directions. If $\mathbf{x}_b$ represents the co-ordinates on the second cable and l_b is the length of a segment in the second cable, we can write the optimization problem as follows:

$$\underset{\mathbf{x}_0,\mathbf{x}_a,\mathbf{x}_b}{\arg\min} \quad \sum_{i=0}^{n-1}\left[\left(\arccos\left(\frac{\mathbf{x}_0^{i+1}-\mathbf{x}_a^{i+1}}{\|\mathbf{x}_0^{i+1}-\mathbf{x}_a^{i+1}\|}\cdot\frac{\mathbf{x}_0^{i}-\mathbf{x}_a^{i}}{\|\mathbf{x}_0^{i}-\mathbf{x}_a^{i}\|}\right)\right)^2+\left(\arccos\left(\frac{\mathbf{x}_0^{i+1}-\mathbf{x}_b^{i+1}}{\|\mathbf{x}_0^{i+1}-\mathbf{x}_b^{i+1}\|}\cdot\frac{\mathbf{x}_0^{i}-\mathbf{x}_b^{i}}{\|\mathbf{x}_0^{i}-\mathbf{x}_b^{i}\|}\right)\right)^2\right]$$

Subject to:

$$\begin{aligned}
&\|\mathbf{x}_0^i-\mathbf{x}_0^{i-1}\| = l_0, \quad \|\mathbf{x}_a^i-\mathbf{x}_a^{i-1}\| = l_a, \quad \|\mathbf{x}_b^i-\mathbf{x}_b^{i-1}\| = l_b\\
&\quad \|\mathbf{x}_0^i-\mathbf{x}_b^i\| = a, \quad \|\mathbf{x}_0^i-\mathbf{x}_a^i\| = a,\\
&\arccos\left(\frac{\mathbf{x}_a^i-\mathbf{x}_0^i}{\|\mathbf{x}_a^i-\mathbf{x}_0^i\|}\cdot\frac{\mathbf{x}_b^i-\mathbf{x}_0^i}{\|\mathbf{x}_b^i-\mathbf{x}_0^i\|}\right) = \arccos\left(\frac{\mathbf{X}_a^i-\mathbf{X}_0^i}{\|\mathbf{X}_a^i-\mathbf{X}_0^i\|}\cdot\frac{\mathbf{X}_b^i-\mathbf{X}_0^i}{\|\mathbf{X}_b^i-\mathbf{X}_0^i\|}\right)\\
&\mathbf{f}(\mathbf{x}_0)\succeq 0, \quad \mathbf{f}(\mathbf{x}_a)\succeq 0, \quad \mathbf{f}(\mathbf{x}_b)\succeq 0\\
&\quad i = 1,2,3,\ldots,n
\end{aligned} \tag{11}$$

Given data: $\mathbf{X}_0, \mathbf{X}_a, \mathbf{X}_b, l_0, l_a, l_b, a, \mathbf{f}(\mathbf{x})$

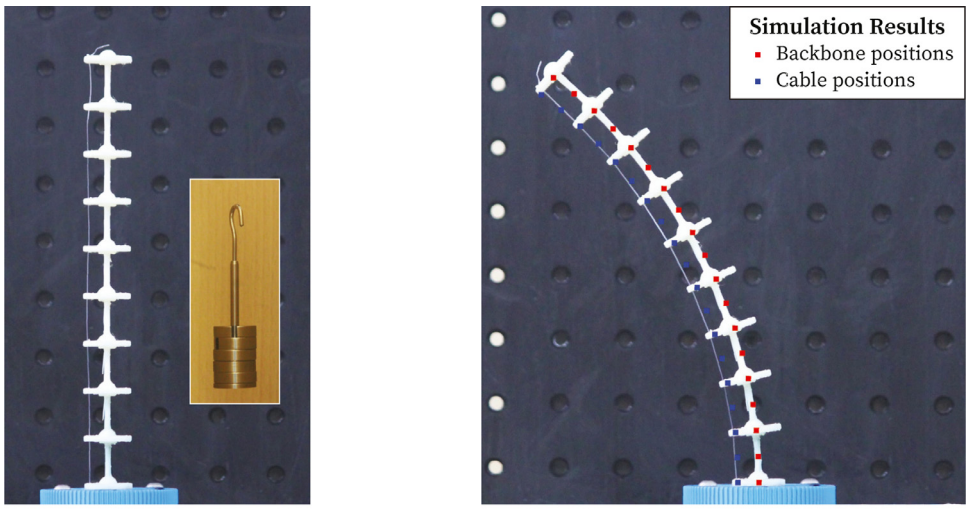
The extra constraint on the angles between the vectors $\mathbf{x}_a$ and $\mathbf{x}_b$ ensure that the position of the holes remain fixed during bending.

3. Results and discussion

To validate the formulation, the equations are numerically solved and compared with the profile of a robot prototype. The prototype (Fig. 4(a)) is 3D printed using ABS material and is 183 mm long with 10 spacer disks of 3 mm each attached at equal spacing.

This fixes the length l_0 as 20 mm. The disks have 12 holes of 1.3 mm diameter arranged around the periphery of a circle with 8 mm radius. Hence, $a = 8$ mm for this robot. For the cable, nylon cord of diameter 0.5 mm is used and is attached to the topmost disk. The deformation of the robot is captured using a camera positioned perpendicular to the plane of bending, and the profile obtained from theoretical formulation is super-imposed on the captured images for comparison. Fig. 4(b) shows the actual deformation of the robot prototype with the simulated results superimposed on the same.

The red markers show the values of $\mathbf{x}_0$ and the blue markers show the values of $\mathbf{x}_a$. The optimization problem was solved using `fmincon` routine in MATLAB considering 18 segments (19 nodal co-ordinates) and interior-point algorithm specified in the algorithm option. The co-ordinates of nodes of the backbone and the co-ordinates of the nodes on the cable are assigned to two separate vector arrays in a main script. The initial values of nodes are taken such that the robot



(a) (b)

Fig. 4. (a) Fabricated prototype of CCR with single cable passing through the disks. The cable is terminated at the topmost disk. The CCR is loaded by hanging known weights from below, as shown in the inset. (b) CCR bending with simulated results superimposed on experimental photograph.

is initially vertical. One of the arguments of `fmincon` routine is the function to be minimized, which is calculated from the first expression in equation (4). The optimization routine also collects the equality constraints in the form of a single vector function, as a separate argument. A main script runs the `fmincon` routine by progressively reducing the value of l for each iteration. The initial values of nodal vectors for each iteration is chosen as the final values obtained from the previous iteration so as to reduce the computation time. A total of 76 optimization variables are solved at once which took about 2.6 seconds per iteration in an Intel processor at 2.00 GHz and 8 GB RAM. For all the experiments, the cable is pulled by a certain amount, and the reduction of lengths of each cable segments are assumed to be equal, i.e., $l_a = l_0 - \frac{l}{n}$ for all the segments. Cable displacement is computed from photos using image processing. The root mean square of the difference between the value of $\mathbf{x}_0$ obtained experimentally and numerically is used as a measure of accuracy of estimation. For the results shown in Fig. 4(b), the RMS error is 1.35 mm, which is 0.75% of the length of the CCR.

3.1. Experimental validation of kinematics with obstacles

The solution to the optimization problem (5) is compared with experimental results for a single obstacle in 2D plane and is shown in Fig. 5(a).

An object with circular cross-section of diameter 30 mm is fixed at a distance of 44 mm in the X direction and 104 mm in the Y direction from the base. The proposed formulation correctly predicts the pose of the CCR with an error of 0.58% of the total length of the CCR. In Fig. 5(b), the numerical solution is compared with experimental profile for CCR in contact with two obstacles. The first obstacle of diameter 25 mm is kept at a distance of 33 mm and 89 mm from the base in the X and Y directions respectively, the second obstacle of diameter 30 mm is kept at 59 mm and 140 mm from the base in the X and Y directions respectively. The error in this case in 0.58% of the total length of the CCR. In both cases, computation time is about 4 seconds.

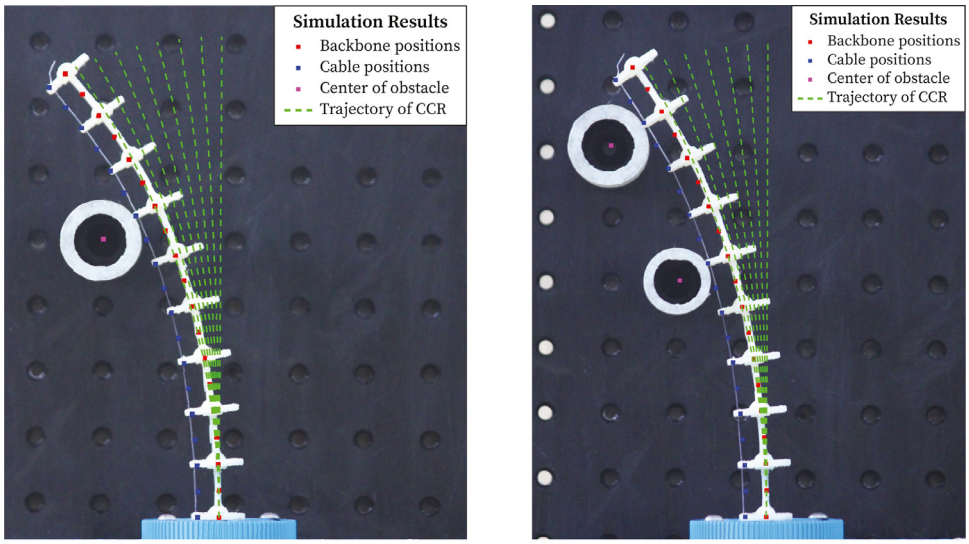
The Lagrange multipliers corresponding to the inequality constraint can be obtained from the output of the `fmincon` routine. Fig. 6 show the Lagrange multipliers corresponding to the inequality constraints $f_1(\mathbf{x}_a) \succeq 0$ and $f_2(\mathbf{x}_a) \succeq 0$, respectively. The peak of the curves occur at the nodes 11 and 17 which corresponds to the disks 6 and 9 from the bottom. These are the actual contact locations as can be observed from Fig. 5(b).

3.2. Experimental validation of contact force measurement

The values for the tip displacement in x, y and θ directions as functions of l can be obtained from the kinematic analysis as shown in Fig. 7.

Linear curves are fitted in this profile whose slopes directly give the values of $f_x', f_y',$ and f_θ'. Fig. 8 shows the experimental set-up where a load cell is kept at 26 mm from the backbone in the X direction and 127 mm from the base.

As the CCR is pulled, the third disk from the top comes into contact with the load cell and the force readings in X direction are directly observed from the meter. The datum for the applied load F_l is set to the value where the CCR is in contact with the load cell. As per the experimental set-up, the component F_y should be zero since the direction is not constrained and the tip is allowed to slide along the surface of the load cell. However, this is not the case due to friction



(a) (b)

Fig. 5. Resulting deformation for CCR in contact with (a) one obstacle, (b) two obstacles.

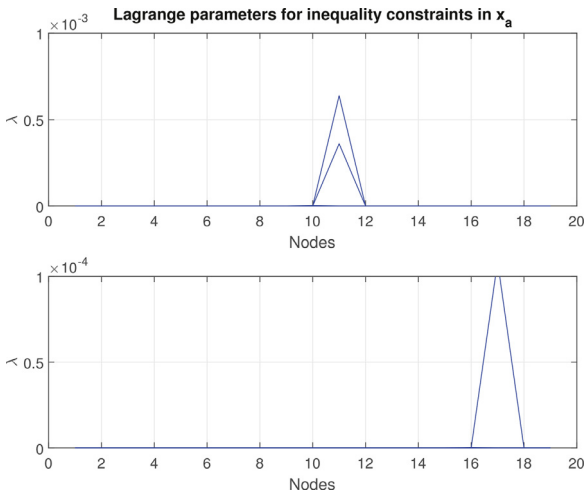


Fig. 6. Contact position estimation from Lagrange multipliers.

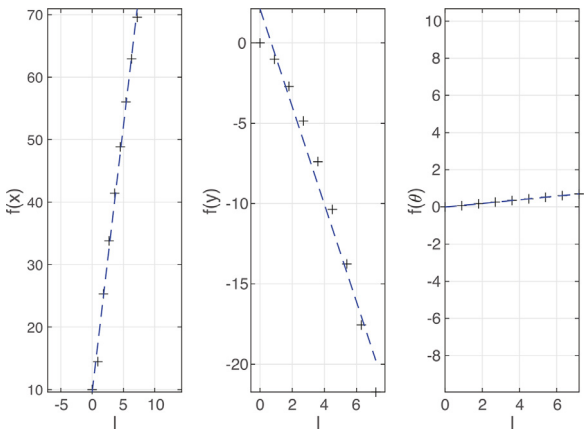


Fig. 7. Tip displacement in X, Y, θ directions as function of cable displacement.

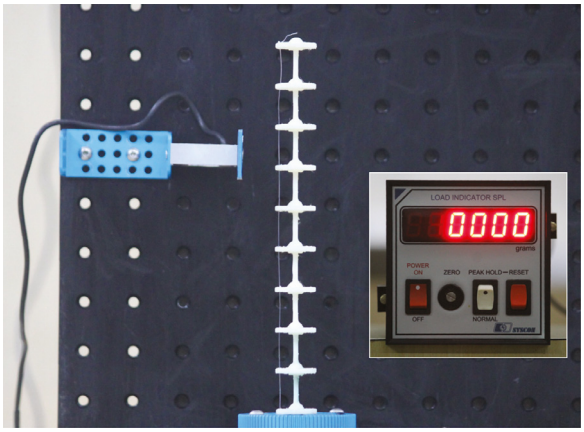


Fig. 8. Experimental set-up for measuring contact forces. Force value is obtained from the meter in grams as shown in the inset.

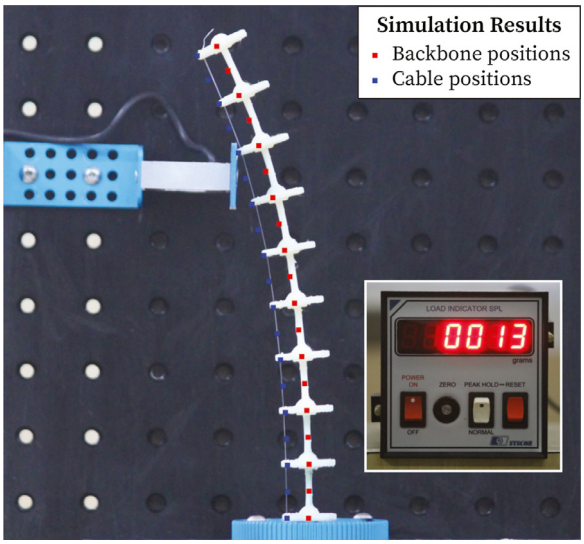


Fig. 9. Contact force and kinematics for obstacle contact at third disk from the top.

between the load cell and the CCR. In order to account for this force, we assume the vertical force $P_y = \mu P_x$. Then from equation (10), we get

$$P_x = -\frac{aF_l f'_\theta}{\mu f'_y + f'_x} \tag{12}$$

The quantity μ is found out from one experimental data point and is used throughout the rest of this paper.

Fig. 9 shows the experimental results with the numerical results superimposed on the figure.

Since the load cell imposes constraint in the X-direction only on the 15th node, the constraint equation is prescribed as

$$\mathbf{f}(x) = d - x_a^{[15]} \tag{13}$$

where $x_a^{[15]}$ is the X-component of the position vector $\mathbf{x}_a$ at the 15th node and d is the distance between the backbone and the load cell. The values of forces obtained experimentally is plotted alongside the numerical solution in Fig. 10(a).

The experimental results accurately match the theoretical results with maximum error less than 1.0 g – the least count of the load cell used for force measurements. More experiments are conducted by varying the distance between the backbone and the load cell and the comparison between experimental and theoretical values are tabulated in Table 2[2].

[2] Each measurements were repeated 6 times and the mean value, rounded off to the nearest integer, is shown in the Table.

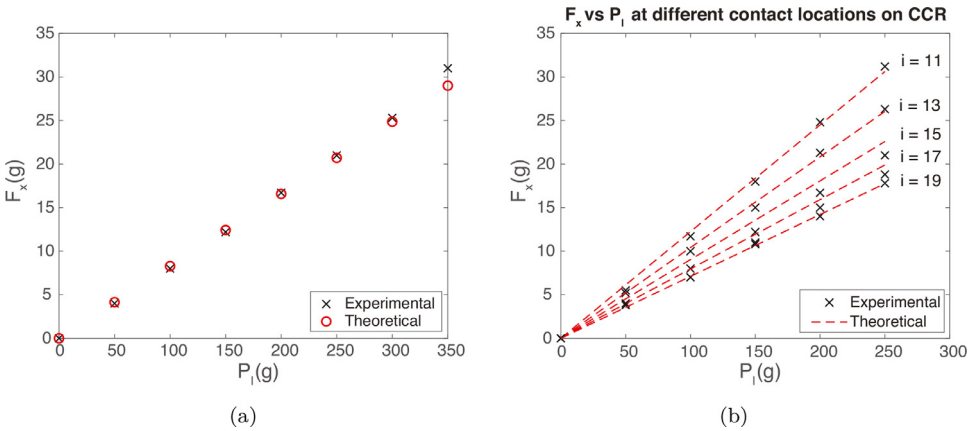


Fig. 10. Comparison between experimental and theoretical values of contact forces (a) when the third disk from the top touches the load cell, (b) for different contact points in CCR.

Table 2
Experimental vs theoretical values of contact force when load cell is placed at d distance from the centre of backbone.

F_l (gram)	Experimental F_x (gram)				Theoretical F_x (gram)
	$d = 24$ mm	$d = 31$ mm	$d = 36$ mm	$d = 51$ mm	
50	3	4	3	3	3.5
100	7	7	7	7	7.1
150	10	11	10	10	10.7
200	14	14	14	14	14.2
250	17	18	17	17	17.8
300	20	21	21	21	21.3
350	23	25	24	24	24.9

From experiments, the value of force is observed not to appreciably change with offset d. Further experiments are conducted by changing the point of contact on the CCR. The resulting curves are plotted in Fig. 10(b) for comparison. For all cases, the average error in prediction is less than the least count of 1.0 g.

3.3. Simulation of kinematics with obstacles in 3D

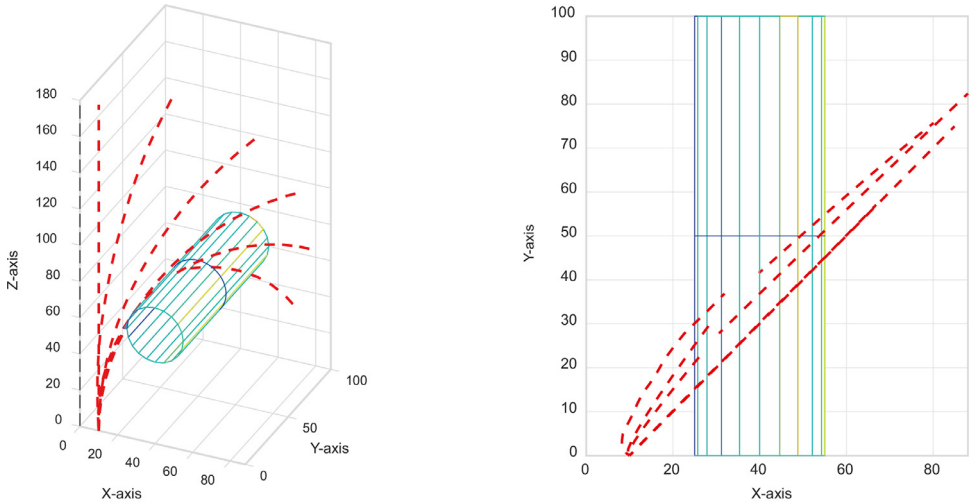
Simulated results for the bending of CCR in 3D is shown in Fig. 11. Accurate measurement of the CCR pose in 3D as well as estimating the three dimensional contact forces to validate the theoretical solution is a work currently underway.

3.4. Discussion

The developed framework is able to predict the pose of the CCR with maximum error of 0.75% of the total length of the CCR. All the numerical procedures are conducted by gradually increasing the cable contraction through small incremental steps. As expected, the convergence of problems with obstacles is slower compared to those without obstacles. However, the time taken for convergence for a single obstacle and two obstacles is almost the same. For the `fmincon` procedure presented in this paper, the authors did not manually specify the gradient of objective function. We believe that by specifying the gradient, the convergence time could be further reduced.

The deviation between theoretical and experimental results for static analysis could be due to measurement errors from the experimental set-up. Possible sources of errors are cable extensions under large weights, friction between the discs and cable and dimensional inaccuracy arising from the surface finish of the 3D printed prototype. It should also be mentioned that the proposed method for static analysis does not take into account the material properties of the CCR and also does not derive its equations from elastic equilibrium equations. Hence, it is not possible to know the stress state of the system using this method, which is one of its disadvantages.

As mentioned in Section 1 of the paper, there exists many methods in the literature which are shown to calculate the kinematics of the robot with good degree of accuracy. However, the method presented in this paper is experimentally verified to give accurate results while being methodologically simpler. This is because calculating the post-collision behaviour of the CCR using available theoretical models require the knowledge of exact contact location on the CCR–which is either directly obtained from observations or estimated using advanced controllers with the help of measured tip-pose data. In



(a) Isometric view (b) Top view

Fig. 11. Trajectory of CCR in 3D where two cables are actuated by equal amounts.

the proposed model however, prescription of the contact location itself is not necessary for calculating the post-collision kinematics. A single formulation captures the characteristics for both pre-collision and post-collision kinematics, which is the major advantage of this model.

4. Conclusions and future work

In this paper, an optimization-based framework for the kinematics of cable-driven continuum robots is presented. The framework is derived from the minimum strain-energy principle and is experimentally shown to be accurate for predicting the deformed pose of a robot with 0.75% error. In the developed method, presence of obstacles can be directly incorporated in the optimization problem as inequality constraints. The method shown in the paper requires only the knowledge of the position and shape of the obstacle, instead of the exact location of contact, to calculate the profile of the robot after collision. It is also shown that the contact location can be obtained by observing the Lagrange multipliers of the inequality constraints in the optimization problem. The solution from experiments show that the framework gives the pose of a CCR with an error of 0.58% in presence of single and multiple obstacles. A simplified approach to estimate the contact forces using the derived kinematics and the method of virtual work is demonstrated. The statics analysis is shown to estimate the contact forces with average error of less than 1.0 g.

As shown, the kinematic model can be extended to 3D motion of the CCR when two cables are actuated. Development of 3D force measurement for the 3D contact is a work in progress. Improving the model to reduce the computation time for real-time implementation in control systems is also a work in progress.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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